

CHAPTER 2

2.0 Customer

Introduction

The entities required for the Duke Bike Showroom Management System are identified based on the business requirements of the organization. The attributes of each entity, along with their primary keys, foreign keys, and data types, are defined to ensure proper data organization and efficient data management. The relationships between the entities are identified to represent the interactions within the system, including customer management, bike management, dealer management, inventory management, order processing, and payment management. In addition, the cardinality of each relationship is determined to define how the entities are associated with one another. This analysis provides a clear understanding of the database requirements and establishes a strong foundation for designing an efficient relational database while ensuring data integrity, consistency, and minimal redundancy.

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique customer ID
First_Name	VARCHAR(50)	-	Customer first name
Last_Name	VARCHAR(50)	-	Customer last name
Gender	VARCHAR(10)	-	Gender
Date_of_Birth	DATE	-	Date of birth
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Mobile number
Address	TEXT	-	Customer address
City	VARCHAR(50)	-	City
State	VARCHAR(50)	-	State
Pincode	INT	-	Postal code
Registration_Date	DATE	-	Registration date
License_Number	VARCHAR(20)	-	Driving license number

2.1 Bike

Attribute	Data Type	Key	Description
Bike_ID	INT	PK	Unique bike ID

Model_Name	VARCHAR(100)	-	e.g. Duke 200, Duke 250, Duke 390
Category_ID	INT	FK	References Category
Dealer_ID	INT	FK	References Dealer
Price	DECIMAL(10,2)	-	Ex-showroom price
Engine_CC	INT	-	Engine displacement in cc
Color	VARCHAR(30)	-	Available color
Fuel_Type	VARCHAR(20)	-	Petrol/Electric
Brand	VARCHAR(50)	-	Brand name (KTM)
Manufacture_Year	INT	-	Year of manufacture

2.2 Category

Attribute	Data Type	Key	Description
Category_ID	INT	PK	Unique category ID
Category_Name	VARCHAR(50)	-	e.g. Naked, Adventure, Sports
Category_Description	TEXT	-	Category description
Status	VARCHAR(20)	-	Active / Inactive

2.3 Dealer

Attribute	Data Type	Key	Description
Dealer_ID	INT	PK	Unique dealer ID
Dealer_Name	VARCHAR(100)	-	Dealer / showroom name
Contact_Person	VARCHAR(50)	-	Contact person
Phone	VARCHAR(15)	-	Phone number
Email	VARCHAR(100)	-	Email address
City	VARCHAR(50)	-	Dealer city
GST_Number	VARCHAR(20)	-	GST registration number

2.4 Inventory

Attribute	Data Type	Key	Description
Inventory_ID	INT	PK	Unique inventory ID

Bike_ID	INT	FK	References Bike
Quantity	INT	-	Available quantity
Minimum_Stock	INT	-	Minimum stock level
Showroom_Location	VARCHAR(100)	-	Showroom / warehouse location
Last_Updated	DATE	-	Last updated date

2.5 Orders

Attribute	Data Type	Key	Description
Order_ID	INT	PK	Unique order ID
Customer_ID	INT	FK	References Customer
Order_Date	DATE	-	Order date
Total_Amount	DECIMAL(10,2)	-	Total amount
Order_Status	VARCHAR(30)	-	Pending/Booked/Delivered
Delivery_Address	TEXT	-	Delivery address
Delivery_Date	DATE	-	Expected delivery date

2.6 Order Details

Attribute	Data Type	Key	Description
Order_Detail_ID	INT	PK	Unique order detail ID
Order_ID	INT	FK	References Orders
Bike_ID	INT	FK	References Bike
Quantity	INT	-	Ordered quantity
Unit_Price	DECIMAL(10,2)	-	Price per bike
Discount	DECIMAL(10,2)	-	Discount amount
Total_Price	DECIMAL(10,2)	-	Total price

2.7 Payment

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK	References Orders
Payment_Method	VARCHAR(30)	-	UPI/Card/Loan/COD

Transaction_ID	VARCHAR(100)	-	Transaction reference
Payment_Status	VARCHAR(20)	-	Success/Pending
Payment_Date	DATE	-	Payment date
Amount	DECIMAL(10,2)	-	Paid amount

2.8 Admin

Attribute	Data Type	Key	Description
Admin_ID	INT	PK	Unique admin ID
Admin_Name	VARCHAR(100)	-	Administrator name
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Contact number
Role	VARCHAR(30)	-	Admin role
Username	VARCHAR(50)	-	Login username
Password	VARCHAR(100)	-	Login password

2.9 Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Orders	1 : M
Orders	Contains	Order Details	1 : M
Bike	Appears In	Order Details	1 : M
Category	Contains	Bike	1 : M
Dealer	Supplies	Bike	1 : M
Bike	Stored In	Inventory	1 : 1
Orders	Has	Payment	1 : 1
Admin	Manages	Bike	1 : M
Admin	Manages	Orders	1 : M